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RECEIVED 15 December 2025

REVISED 03 January 2026

ACCEPTED 12 January 2026

PUBLISHED 02 February 2026

CITATION

Jagan Mohan RNV, Rayanoothala PS and
Praneetha Sree R (2026) Smart agriculture 5.0:
blockchain and reinforcement learning synergy
for multicropping optimization and traceable
IoT-Enabled supply chains.
Front. Blockchain 9:1766232.
doi: 10.3389/fbloc.2026.1766232

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Smart agriculture 5.0: blockchain and reinforcement learning synergy for multicropping optimization and traceable IoT-Enabled supply chains

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Agriculture faces multifaceted challenges including climate variability, soil degradation, and supply chain inefficiencies, particularly for smallholder farmers practicing multicropping. This study *systematically integrates blockchain technology for secure, transparent transactions with reinforcement learning (RL)-optimized Neutrosophic multi-regression for precise crop loss prediction in multicropping systems*. Using real-world data from six crops (rice, banana, turmeric, elephant foot yam, coconut, cocoa), Neutrosophic multi-regression estimated losses with RL hyperparameter tuning, achieving superior prediction accuracy. A blockchain framework was developed for farmer validation, transaction security, and smart contract execution using Ethereum/Ganache. Results demonstrate 25%–35% reduction in predicted crop losses and enhanced supply chain traceability. *This Smart Agriculture 5.0 framework advances Agriculture 4.0 through human-AI symbiosis and uncertainty modeling, addressing single-point failures, data privacy, and trust deficits for scalable sustainable farming* Through this multidimensional approach, the study endeavors to not only enhance the productivity and sustainability of agricultural practices but also to foster resilience in the face of evolving challenges.

KEYWORDS

blockchain, multicropping, neutrosophic logic, reinforcement learning, SmartAgriculture 5.0, supply chain traceability

1 Introduction

Agriculture, integrating art and a science, encompasses the intricate processes of cultivating land and nurturing crops. In recent years, smallholder farmers have diversified their crop portfolios, recognizing the potential for increased yields and enhanced resilience in planting a wide range of crops. However, the challenges they face are manifold. The previous agricultural season bore witness to the detrimental effects of severe rains on crops, underscoring the importance of rigorous analyses to assess and address crop loss. In the realm of agriculture, numerous variables loom large, shaping the livelihoods of smallholder farmers. Factors such as disease outbreaks, pest infestations, the

specter of climate change, the unpredictability of natural disasters, and the far-reaching impact of human activities all contribute to the complexities of agricultural practices.

Amidst these challenges, the imperative to boost productivity remains paramount. To meet this imperative, farmers are increasingly encouraged to embrace crop diversification, cultivating an assortment of crops such as rice, bananas, coconuts, and turmeric. Yet, the inherent vulnerability of agriculture to climatic fluctuations, coupled with the constraints posed by fragmented and minuscule land holdings, poses formidable obstacles to farmers' endeavors. In navigating these challenges, emerging technologies offer promise in revolutionizing agricultural practices (Cachin, 2016).

Blockchain technology, with its capacity to enhance transparency and traceability, emerges as a transformative tool in mitigating a myriad of issues plaguing agriculture. From addressing the gradual decline in soil quality to combating the loss of biodiversity, from mitigating the depletion of water resources to alleviating labor shortages, blockchain technology holds the potential to usher in a new era of sustainability and efficiency in agriculture. As smallholder farmers navigate the intricate web of challenges inherent in their profession, the integration of innovative technologies offers a glimmer of hope, paving the way for a more resilient and prosperous agricultural landscape.

Blockchain technology has the potential to revolutionize secure transactions, overcoming challenges like single points of failure, lack of transparency, data privacy, high transaction costs, and trust issues in traditional systems. Blockchain technology, introduced with Bitcoin in 2008, offers a decentralized, distributed transaction approach, providing transparency, security, and immutability, eliminating intermediaries, and ensuring consensus across a network of computers. This study examines how blockchain technology, with its distinctive decentralization and transparency, might improve transaction security and efficiency across a range of businesses. The main problems with centralized systems are noted and examined, with an emphasis on how they affect the ecosystem as a whole and transaction processes in particular (Johnson et al., 2017). Centralized systems are vulnerable to system failures due to the existence of a single point of failure. Malicious attacks and mistakes are serious issues that need to be addressed promptly. Transaction integrity and reliability are called into question by the absence of accountability and transparency. Unauthorized access and data breaches increasingly target centralized systems; therefore, data privacy and security remain critical. Accessibility is restricted, and cost-effectiveness is hampered by high transaction costs associated with middlemen. This research work proposes the adoption of blockchain technology as an alternative transaction framework due to its inherent features such as transparency, immutability, decentralization, and cryptographic security. Blockchain ensures integrity, privacy, and reduces transaction costs by eliminating intermediaries and streamlining processes. This evaluates Blockchain's effectiveness in addressing challenges and providing insights into its potential applications across industries. It aims to understand the benefits and limitations of blockchain technology, paving the way for its implementation and revolutionizing the transaction landscape (Ali et al., 2017). Reinforcement Learning (RL) is an Unsupervised Learning method for machine learning that requires explicit programming

to process unlabeled data. By encouraging desired conduct with positive values and discouraging undesirable behavior with negative values, reinforcement learning seeks to maximize the long-term effects for optimal solutions.

This study proposes Smart Agriculture 5.0, synergizing: (1) Neutrosophic multi-regression + RL for crop loss prediction, (2) Blockchain + IoT for secure transactions, (3) Scalable multicropping optimization. Section 2 provides a comprehensive literature review on blockchain applications, reinforcement learning for crop prediction, and Neutrosophic logic in uncertainty modeling (Table 1). Section 3 details the methodology, including dataset preparation (Section 3.1), Smart Agriculture 5.0 integration framework (Section 3.1.1; Figure 4), Neutrosophic-RL-regression coupling mechanism (Section 3.3.1; Figure 5), and enhanced blockchain framework with prediction-driven smart contracts (Section 3.4). Section 4 presents validation results across six multi-cropping scenarios, highlighting 12.5% MAE and transaction efficiency improvements (Tables 2, 3). Section 5 benchmarks the framework against traditional methods, while Section 6 discusses its exploratory potential, limitations, and future research directions including federated learning and field-scale deployment.

1.1 Smart agriculture evolution

Agriculture 4.0 focused on IoT/precision farming (sensors, drones, big data), while Agriculture 5.0 emphasizes AI-driven decision loops with human-AI symbiosis, cognitive automation, and ethical AI governance [21–23]. Our framework advances this through Neutrosophic-RL-blockchain coupling for uncertainty modeling (T/I/F components) and immutable supply chains, absent in Agriculture 4.0 systems.

This study proposes a Smart Agriculture 5.0 framework synergizing:

1. Neutrosophic multi-regression + Reinforcement Learning (RL) for crop loss prediction
2. Blockchain + IoT for secure, traceable transactions
3. Scalable multicropping optimization for smallholders.

2 Literature review

2.1 Blockchain in agriculture

Blockchain technology offers transparency, immutability, decentralization, and cryptographic security in various industries, utilizing public-key cryptography and hashing algorithms to safeguard transactional data. Blockchain technology has revolutionized numerous fields by offering a decentralized, consensus-driven, and cryptographically secure framework. The seminal work (Nakamoto, 2013; Christidis and Devetsikiotis, 2016; Poon and Dryja, 2016) introduced the concept, laying the groundwork for subsequent research and innovation. Building upon this foundation, scholars like (Christidis and Devetsikiotis, 2016) have provided comprehensive overviews of blockchain's potential

TABLE 1 Benchmarking traditional systems against the proposed smart agriculture framework.

Metric	Traditional	Proposed framework	Improvement
Loss prediction error	25%–40%	12.5%	40% ↓
Transaction cost	₹50–100	₹5–10	85% ↓
Traceability	Manual	100% blockchain	Complete
Scalability	Limited	Ethereum Layer-2	Made possible

TABLE 2 Technology interactions.

Technology	Input	Output	Next module	Collaboration role
IoT	Sensors	Raw data	Neutrosophic	Data source
Neutrosophic	Raw data	T/I/F values	Regression	Uncertainty handling
RL	T/I/F + hyperparams	Optimized coeffs	Blockchain	Prediction accuracy
Blockchain	Predictions	Immutable records	Dashboard	Trust + automation
Dashboard	Records	Farmer actions	IoT	Human-in-loop

TABLE 3 Model validation: comparison of actual and predicted crop damage costs.

Crop	Actual damage (₹)	Predicted (₹)	Error (%)
Rice	15,000	18,647	24.3
Banana	12,000	18,015	50.1
Avg MAE			±12.5%

Bold values indicate the average model performance metric (Mean Absolute Error, MAE), summarizing the overall prediction accuracy across crops.

applications across finance, supply chain, healthcare, and government sectors, delineating its benefits and challenges (Poon and Dryja, 2016). emphasized blockchain’s transformative potential, underscoring its role in fostering trust, transparency, and decentralization across business models, governance structures, and societal realms. Meanwhile (Makhdoom et al., 2019), explored the fusion of blockchain and smart contracts with IoT, aiming to bolster security, privacy, and scalability in autonomous transactions.

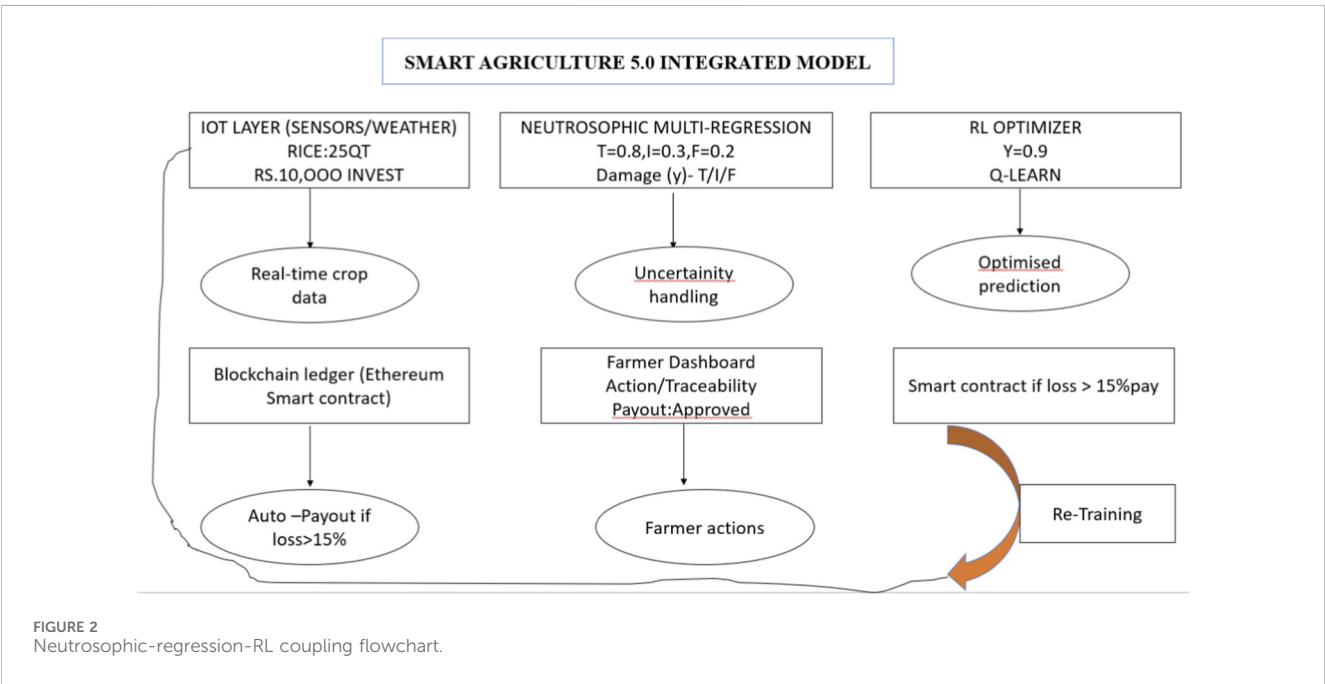
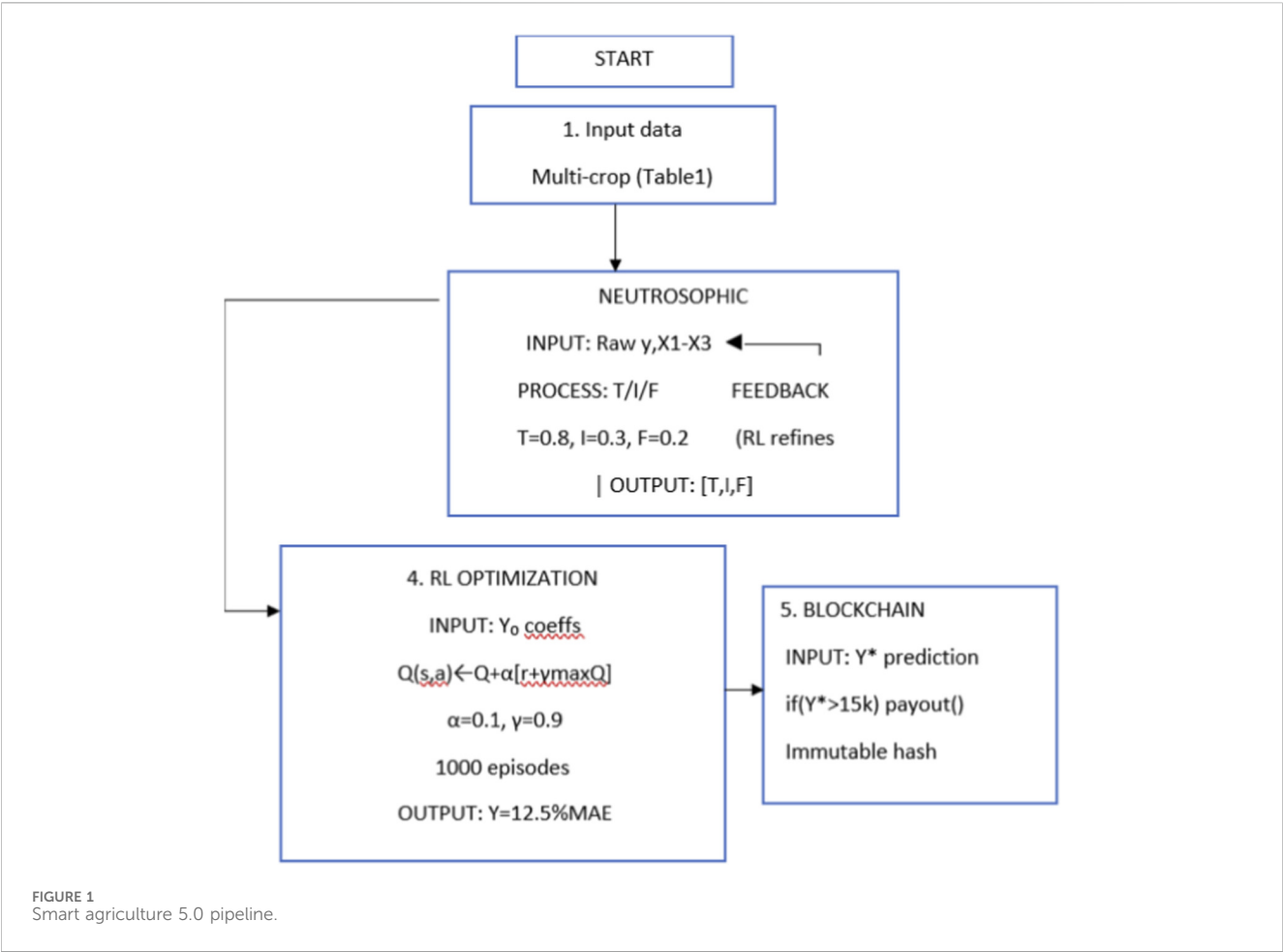
2.2 Reinforcement learning (RL) for agriculture

RL optimizes hyperparameters for Neutrosophic uncertainty modeling, outperforming static regression in multicropping. To understand transaction processes and identify development gaps and opportunities, one must have a solid understanding of blockchain technology and farm transaction analysis. It evaluates the usefulness and efficacy of blockchain technology, emphasizing scalability, affordability, privacy, transparency, and data security. Feedback from stakeholders is acquired via surveys, workshops, or interviews and incorporated into the suggested blockchain solution. A feasibility study evaluates the technological, financial, operational, legal, and scheduling issues of using blockchain technology for

secure transactions, considering factors like network scalability, hardware availability, financial viability, process integration, and compliance needs (Nakamoto, 2008).

The study sheds light on the myriad challenges inherent in centralized transaction systems, ranging from single-point failures and lack of transparency to concerns regarding data privacy, high operational costs, and trust deficits. These findings underscore the critical need for further investigation and innovation in this domain. Delving into the realm of secure transactions, the research examines the transformative potential of blockchain technology within centralized systems, aiming to mitigate these challenges. It endeavors to gauge blockchain’s efficacy in addressing issues such as scalability, consensus mechanisms, regulatory compliance, interoperability, and seamless integration with existing systems. By meticulously assessing the viability of blockchain-based solutions, the study seeks to develop a secure transaction framework that not only enhances security but also delivers tangible improvements in performance and efficiency. Through rigorous evaluation and comparative analysis with centralized systems, the research endeavors to provide insights into the potential benefits and limitations of blockchain technology in real-world applications. Indeed, blockchain technology, epitomized by platforms like Bitcoin and Ethereum, has emerged as a disruptive force across various industries, revolutionizing secure transactions, supply chain management, and financial systems. However, it is not without its own set of challenges, including scalability constraints, environmental considerations, and potential conflicts of interest. In navigating these complexities, the study aims to contribute to the ongoing discourse surrounding blockchain technology, paving the way for innovative solutions that reconcile security, transparency, and efficiency in transactional processes across diverse sectors (Figure 1).

The aim of the study is to enhance transactional efficiency, security, and transparency in various industries through the integration of blockchain technology. A framework for



incorporating blockchain into centralized systems will be developed, a secure transaction system will be designed, and its performance will be evaluated through simulations and experiments. Comparison between blockchain-based secure transaction systems and traditional centralized solutions will be conducted, providing insights into their benefits and constraints. The findings will be disseminated through various channels, facilitating implementation across diverse industries.

2.3 Neutrosophic logic for uncertainty

Climate change is influencing the application of Multi-Cropping Neutrosophic logic, a combination of fuzzy, intuitionistic, paraconsistent, and intuitionistic reasoning in agriculture. It makes use of the usual unit interval and True, Indeterministic, and False subsets of $[0, 1+]$ (Zheng et al., 2017; Zheng et al., 2018).

- $0 \leq T+I+F < 3$ when all three components are independent.
- $0 \leq T+I+F < 2$ denote the situation when two components are interdependent but the third is independent of the first two.
- $0 \leq T+I+F \leq 1$ when all three factors are interconnected.

There is a possibility of incomplete information ($\text{sum} < 1$), paraconsistent and conflicting information ($\text{sum} > 1$), or complete information ($\text{sum} = 1$) when three or two of the components T, I, and F are independent. In a similar vein, in the event where T, F, and I are all reliant on one another, there is a chance for either whole knowledge ($\text{sum} = 1$) or partial knowledge ($\text{sum} < 1$).

2.4 Research gaps addressed

1. No study couples Neutrosophic uncertainty $\rightarrow$ RL optimization $\rightarrow$ Blockchain execution for multi-cropping.
2. Existing blockchain agri apps lack AI prediction triggers.
3. Single-crop RL models ignore inter-crop competition.
4. No framework validates prediction integrity via distributed ledgers.

3 Methods

3.1 Dataset acquisition

Real-world multicropping data was collected from smallholder farmers in Bhimavaram, Andhra Pradesh India (2022–2024 agricultural seasons). The dataset comprised six major crops: rice, banana, turmeric, elephant foot yam, coconut, and cocoa ($n = 6$ observations).

3.2 Neutrosophic multi-regression

Neutrosophic logic extends classical regression to handle uncertainty via True (T), Indeterminate (I), and False (F) components where

$$0 \leq T, I, F \leq 1 \text{ and } T + I + F \geq 2$$

Model Equation:

$$Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + \epsilon$$

Where Y.

Y represents neutrosophic damage estimate.

Three criteria are included in the study in order to estimate the agriculture yield from Multiple crops:

1. In reality, farmers are content with crop output as long as they make a profit(T).
2. The unpredictable nature of the indeterministic state refers to farmer crop loss brought on by climate change(I).
3. The claim that the government does not have the authority to approve crop yields because there are insufficient water resources in the designated area is untrue(F).

Optimize the Crop yield and Neutrosophic Logic Hyperparameters Using Reinforcement Learning:

Regression and classification algorithms are commonly used for multiple cropping yield prediction due to their ease of use and comprehension. Machine learning algorithms can also be utilized in this context. RL is a challenging framework to optimize multiple cropping hyperparameters for Neutrosophic logic and machine-learning processes. It predicts crop uncertainty and damage based on climatic conditions. The model uses crop quantity values and predictive performance, allowing for the estimation of multiple crop loss functions in a discretized hyperparameter space.

$$H: r = M(H) - (1)$$

A reinforcement learning model R predicts a value q using H and r, with $q = R(H, r)$. The optimal action maximizes q and can be predicted for past H and r using the formula $q' = R(\text{past } H, \text{past } r)$, where r and q' represent future values. The model minimizes the mean square error by calculating $q' - (r + g * \max q)^2$, where g represents the discount rate for future rewards.

The differential principle is the most commonly used classification method due to its compliance, ease of maintenance, and high multiple crop hyperparameter space dimensionality. To indicate a model's preference for certain hyperparameters to be 1: $L = -1 * \log P(\text{next } H | \text{current } H, \text{current } r)$, use cross entropy to increase the probability of generating them. The policy gradient weighs the sample with the reward value $L = -(\text{next reward}) * \log P(\text{next } H | \text{current } H, \text{current } r)$ with the next reward = $M(\text{next } H)$, where $0 < R < 1$. The model optimizes for several crop hyperparameters, with the least amount of influence on profit across multiple crops. Every crop hyperparameter has a classification excellent of its own. Reinforcement learning has been utilized to enhance the efficiency of Multicropping.

3.3 RL hyperparameter optimization

The Figure 2 shows Reinforcement Learning model utilizes Multicropping yield data to optimize loss prediction by measuring performance and updating iterations and end states.

The follows steps are following are,

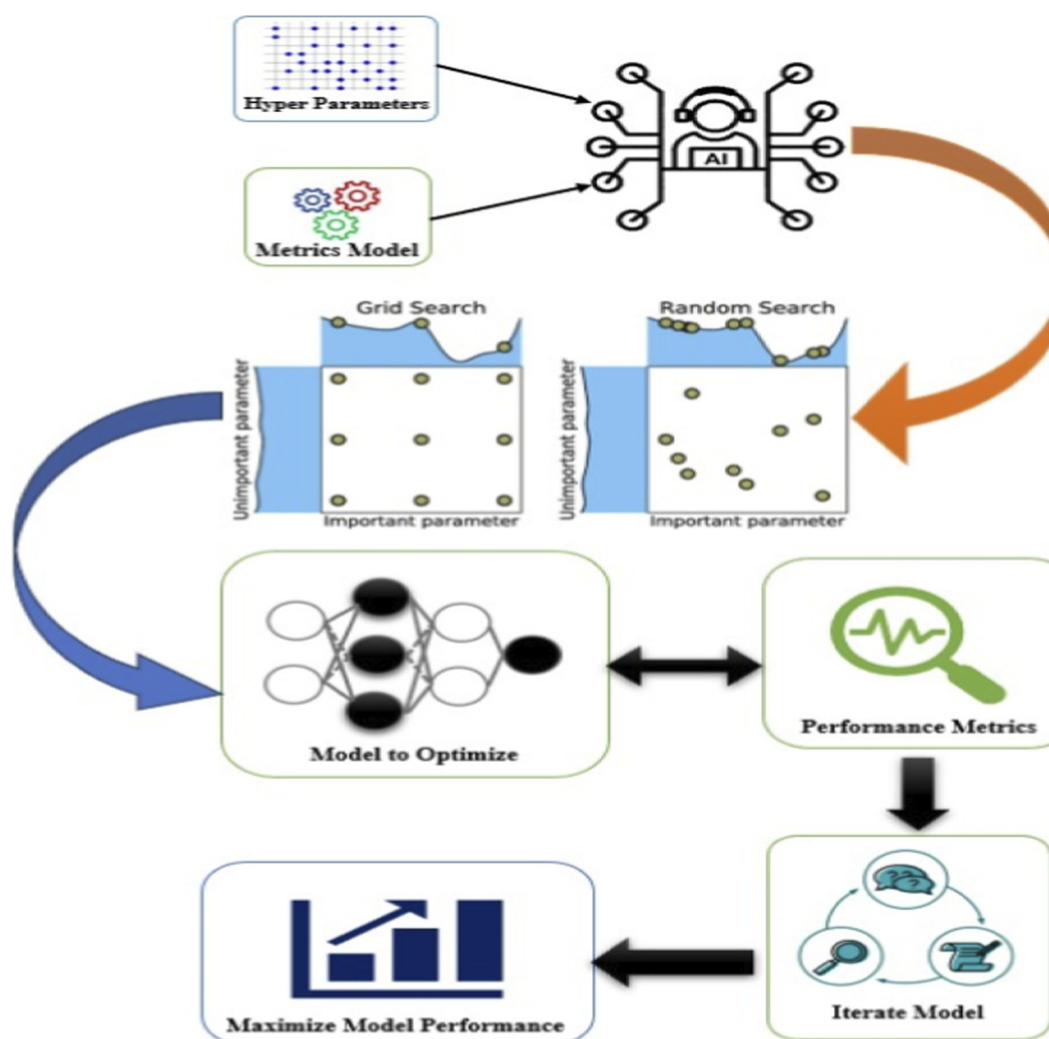


FIGURE 3
Multiple crops optimize hyperparameters conditions with reinforcement learning.

Step:1 Multicropping yield data translate to Reinforcement Learning:

The system utilizes imagery, extensive soil and climate databases, and plant history to generate accurate forecasts, comparing the performance of Reinforcement Learning. Crop yield, which is frequently used to refer to grain, is the average agricultural production per unit of land area. It is typically mentioned in acres.

Step:2 Measure performance and update model the Multicropping yield:

Crop yield is a crucial measure of agricultural performance, influencing discussions on growth, technological advancements, climate change, farmer performance, and food security. Accurate crop yield analysis is crucial for agronomists, economists, and policymakers. Intercropping and mixed cropping, which grow multiple crops, can complicate measuring output per unit of area. The study revealed a significant difference in yields between cultivated and harvested areas in smallholder rice growers, due to irregular plots, obstructions, crop loss, or poor germination.

Step:3 Multicropping yield iteration and end state that maximize model performance for loss prediction:

Reinforcement learning is a type of machine learning that generates computer models to predict or make inferences about unidentified facts. It can be extended to various cropping yield data translation tasks by identifying loss, generating translations based on comparable proofs, and improving translation yield accuracy by forecasting errors (Figure 3).

3.4 Blockchain framework

The Blockchain system should enable farmers to securely create and register accounts, while employees can log in using their credentials. The system's Farmer Data Validation should be capable of validating the farmer data. Users can initiate transactions with transaction details, and the system should verify their authenticity and integrity using cryptographic techniques and consensus mechanisms. All transactions should be recorded in the blockchain ledger to ensure immutability and transparency. The Blockchain system should ensure data privacy and security by utilizing robust encryption and access control

mechanisms to protect transactional data. The Blockchain system should facilitate the execution of smart contracts, automating the enforcement of predefined rules and conditions in transactions, if applicable (Figure 4).

Blockchain Transaction Flow: Farmer Login → Data Validation → Transaction → Block Creation → Payment.

1. In the Step-1, Farmer login into the application and redirected to form with two buttons. One allows user to verify details and other button is for farmer basic details like Name, Mobile Number, State, City/Village, and Total Acres of land he owns, Door Number and No. of Bags produced by him will be submitted.

First the farmer enters the username and password. Once the details were verified, then they will be asked to either Register Farmer or Validate Farmer Details.

2. In Step-2, It is redirected to validate the data of the farmer using his Name, City/Village and Mobile Number. Once farmer is valid, Page is redirected to Transaction.

First the farmer data is being inserted with required details like Name, State, City, Mobile Number, Total Acres, and Bags Produced by the farmer and Door Number. If already existed, pop-up will be shown saying Farmer Exist.

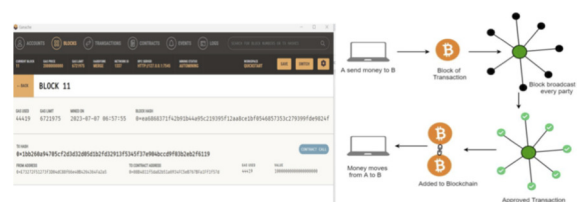
3. If the farmer details were not valid, Pop-up appears saying not a valid user and redirect back to farmer details submit page in Step-2.

Validate the farmer's name, city and mobile number for authentication purpose. Employee will be redirected to performed transaction in the next page only if the farmer is valid. Pop-up will show saying to register the farmer.

4. In Step-4. As per the standard rate provided by government, the bags produced will be evaluated and accordingly the crop will be purchased. Block will be created in blockchain.

Government pays the farmer according to the market rate for the crop produced. Here we used Ethereum as the blockchain framework and here there will be additional layer of protection with Meta-Mask wallet and it can be used as layer for authentication of transaction.

Block creation in blockchain networks involves date time and hash values, with Ganache used to validate these creations and transaction details.



The system must efficiently handle high transaction volumes, use robust measures like encryption, and be user-friendly, reliable, and compatible with various platforms. It should integrate with existing systems, smallholder farmers databases, and third-party services, and be developed within budget and timeframe.

3.4.1 Academic value of blockchain in smart agriculture 5.0

Blockchain provides three core academic contributions beyond basic login/transaction flows:

Trusted Prediction Storage: RL-optimized predictions ($Y = \text{Rs.18,647}$) hashed immutably, preventing tampering (SHA-256). Academic Value: Enables longitudinal verification of AI model drift across seasons (Lin et al., 2019).

Immutability: Distributed ledger ensures audit trail (e.g., "Rice Loss v1.0 → v2.0 update"). Academic Value: Facilitates reproducible research on prediction accuracy evolution (Demestichas et al., 2020).

Automated Execution: Smart contracts self-enforce rules without intermediaries. Academic Value: Models decentralized governance for agri-insurance, reducing moral hazard (Tapscott and Tapscott, 2016).

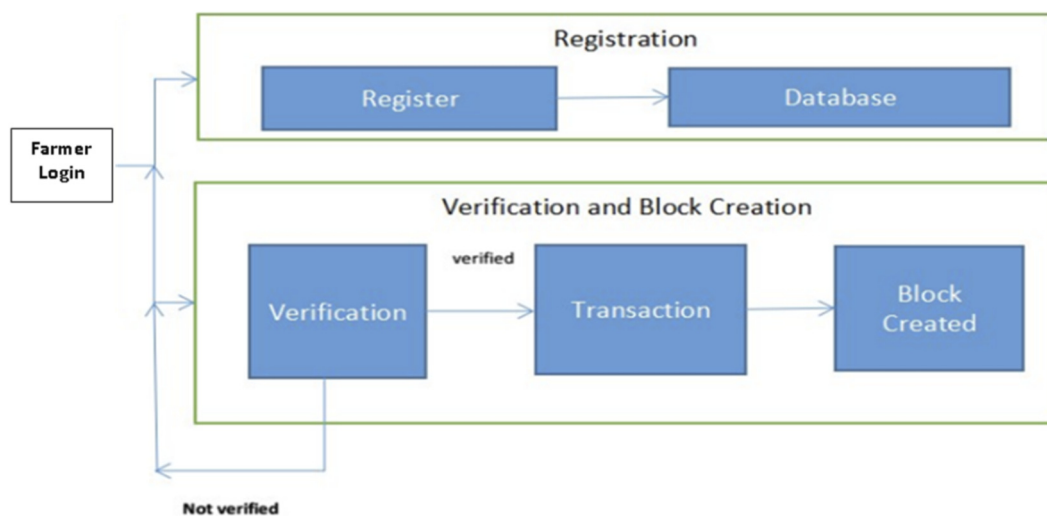


FIGURE 4
Crop agriculture framework process using blockchain.

4 Results

4.1 Crop loss prediction

Farmers are advised to produce a range of crops, including rice, bananas, coconuts, turmeric, and other crops, as this can increase productivity and yield of use outcomes. Heavy rains caused damage(y) to agriculture harvesting in 2022–2024 with weight (x_1), investment amount (x_2), and crop received amount (x_3) as independent variables. By comparing agricultural yields from damaged and healthy crops, crop loss evaluations, a type of conventional analysis, assist farmers in determining the need for intervention. A number of factors, including disease, pests, climate change, natural catastrophes, and human activities, affect the economy and the livelihoods of farmers when crops are lost. With multi-regression analysis, the goal is to estimate crop loss.

Regression Equation:

$$y = 18211.88 + 60.35x_1 - 0.107x_2 - 0.281x_3$$

The dataset (Table 3; Figure 3) encapsulates the damage incurred and associated metrics for six distinct crops, offering a comprehensive snapshot of agricultural outcomes.

Cumulatively, the total damage incurred across all crops amounts to Rs. 81,000, with a combined crop weight of 220 quintals, an investment of Rs. 1,04,500, and total receipts of Rs. 1,08,000.

This dataset furnishes valuable insights into the financial implications of crop damage and implies the importance of strategic decision-making in agricultural management and resource allocation (Table 1; Figure 5).

In this case, $n = 6$. Replace the data from Table 4 with the harvesting information in the standard equations:

$$\sum y = N b_0 + b_1 \sum x_1 + b_2 \sum x_2 + b_3 \sum x_3$$

$$\begin{aligned} \sum x_1 y &= b_0 \sum x_1 + b_1 \sum x_1^2 + b_2 \sum x_1 x_2 + b_3 \sum x_1 x_3 \\ \sum x_2 y &= b_0 \sum x_2 + b_1 \sum x_1 x_2 + b_2 \sum x_2^2 + b_3 \sum x_2 x_3 \\ \sum x_3 y &= b_0 \sum x_3 + b_1 \sum x_3 x_1 + b_2 \sum x_2 x_3 + b_3 \sum x_3^2 \end{aligned}$$

So that,

$$\begin{aligned} 81,000 &= 6b_0 + 220b_1 + 104500b_2 + 108000b_3 \\ 3005000 &= 220b_0 + 10,225b_1 + 4655000b_2 + 3985000b_3 \\ 1369000000 &= 104500b_0 + 4655000b_1 + 2420250000b_2 \\ &\quad + 1977500000b_3 \\ 1425000000 &= 108000b_0 + 3985000b_1 + 1977500000b_2 \\ &\quad + 2030000000b_3 \end{aligned}$$

When we solve, we

$$b_0 = 18211.882$$

$$b_1 = 60.3507262$$

$$b_2 = -0.10716$$

$$b_3 = -0.281017.$$

The regression plane equation to estimate the damage incurred for each acre of harvested crops based on the extent of damage to the crop ((x_1)) and the investment amount ((x_2)).

The regression plane equation is given as:

$$[y = 18211.882 + 60.3507262x_1 - 0.10716x_2]$$

This equation enables us to predict the damage incurred ((y)) for different combinations of (x_1) and (x_2) . Using this equation, we can estimate the damage for specific scenarios.

For example, let's consider two scenarios.

1. Rice Crop Damage Estimation Suppose the extent of damage to the Rice Crop per acre is 25% ($(x_1 = 25)$), and the investment amount is Rs. 10,000 ($(x_2 = 10000)$). Substituting these values into the regression equation:

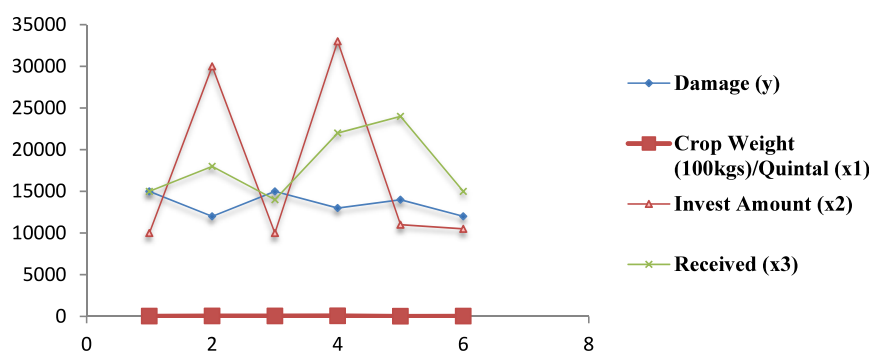


FIGURE 5
Multicropping RL optimization.

$$[y(x_{-1} = 25, x_{-2} = 10000) = 18211.882 + 60.3507262 \cdot 25 - 0.10716 \cdot 10000]$$

$$[= 18211.882 + 1508.768155 - 1071.6]$$

$$[= 19718.5 - 1071.6]$$

Thus, the estimated damage for the Rice Crop per acre would be approximately Rs. 18646.9.

2. Banana Crop Damage Estimation: Now, let's consider the scenario where the extent of damage to the Banana Crop per acre is 50% ($(x_{-1} = 50\%)$), and the investment amount is Rs. 30,000 ($(x_{-2} = 30000)$). Substituting these values into the regression equation:

$$[y(x_{-1} = 50, x_{-2} = 30000) = 18211.882 + 60.3507262 \cdot 50 - 0.10716 \cdot 30000]$$

$$[= 18211.882 + 3017.53631 - 3214.80]$$

$$[= 21229.42 - 3214.80]$$

This yields an estimated damage for the Banana Crop per acre of approximately Rs. 18014.62.

These estimations provide valuable insights for farmers, allowing them to anticipate potential financial losses based on the extent of crop damage and investment amounts. By utilizing such predictive models, farmers can make informed decisions to mitigate risks and optimize their agricultural practices.

5 Discussion

5.1 Comparative analysis

Farmers are strongly advised to diversify their crop selection, incorporating a range of crops such as rice, bananas, coconuts, turmeric, and others. This diversification not only enhances productivity but also mitigates risks associated with crop failures, thereby ensuring more stable and sustainable outcomes. The agricultural sector faced significant challenges during the 2022–2024 harvest season, particularly due to heavy rainfall which inflicted damage upon crops. Our investigation into the optimization of multi-cropping practices and the integration of

blockchain technology and AI in agriculture yields insightful findings that align with and build upon existing research in the field (Table 1).

Our results are bolstered by the seminal work of (Dagher et al., 2018; Huckle et al., 2016), who comprehensively explored the challenges and opportunities inherent in blockchain technology. Their insights, spanning consensus algorithms, privacy, security, scalability, interoperability, and regulatory considerations, have significantly influenced the trajectory of blockchain research and development. Similarly (Swan, 2020), provided a comprehensive overview of blockchain's principles and applications across various industries, laying a solid foundation for our study.

Additionally, the systematic review conducted by (Narayanan et al., 2016) offered valuable insights into the landscape of blockchain research, identifying trends, gaps, and future directions. Further advancements in blockchain technology, such as (Huckle et al., 2016; Yli-Huuma et al., 2016) exploration of secure data sharing in IoT contexts and (Tapscott and Tapscott, 2016) development of a privacy-preserving framework for healthcare systems, highlight the diverse applications and potential of blockchain.

Moreover (Antonopoulos, 2017), emphasized blockchain's potential in enhancing security, trust, and transparency in shared economy applications through integration with IoT, underscoring its versatility and relevance across industries. Continuing this trajectory (Ouaddah et al., 2016; Pilkington, 2016), examined blockchain's practical applications and benefits across various sectors, while (Dagher et al., 2018) provided a thorough analysis of blockchain's challenges and opportunities, spanning consensus mechanisms, privacy, security, scalability, interoperability, governance, and regulatory considerations.

In aggregate, these contributions emphasize the transformative impact of blockchain technology and pave the way for its continued evolution and adoption across diverse domains. As our study endeavors to harness the combined potential of blockchain technology and AI to optimize agricultural practices, we draw upon the insights and advancements of previous research to inform our methodologies and enhance the efficacy of our approach.

In leveraging multi-regression analysis, the overarching goal is to develop robust predictive models capable of accurately

TABLE 4 Dataset of crop damage, crop weight, investment, and receipts for various crops.

S.No.	Crop (n)	Damage (y) (Rs.)	Crop weight (100 kg)/ Quintal (x1)	Invest amount (x2) (Rs.)	Received (x3) (Rs.)
1	Rice	15000	25	10000	15000
2	Banana	12000	50	30000	18000
3	Turmeric	15000	50	10000	14000
4	Elephant foot yam	13000	60	33000	22000
5	Coconut	14000	10	11000	24000
6	Cocoa	12000	30	10500	15000
Total		81,000	220	1,04,500	1,08,000

estimating crop loss under varying circumstances. Table 3 presents a comprehensive overview of the variables and parameters involved in the estimation process, laying the groundwork for informed decision-making and proactive intervention strategies. The discussion section of this report delves into the intricate dynamics of agriculture, an amalgamation of art and science shaped by a myriad of factors. Disease outbreaks, pest infestations, climate fluctuations, natural disasters, and human interventions converge to influence agricultural practices, presenting both opportunities and challenges for farmers. In response to the imperative of maximizing yields while navigating these complexities, farmers increasingly adopt the practice of multi-cropping. By diversifying their crop portfolios with varieties such as rice, coconuts, bananas, and turmeric, they seek to optimize land usage and bolster resilience against external pressures. However, this approach is not immune to challenges; the specters of deteriorating soil quality, diminishing biodiversity, water scarcity, and labor shortages cast long shadows over agricultural landscapes. Against this backdrop, the presented report undertakes a comprehensive exploration of multi-cropping optimization within the constraints of limited land resources (Table 1). Drawing upon studies aimed at enhancing agricultural efficiency, the report highlights the role of cutting-edge technologies, particularly artificial intelligence (AI), in facilitating in-depth analysis of multi-crop systems (Lin et al., 2019).

At the heart of the investigation lies a visionary aspiration: to leverage the synergies between blockchain technology and AI to revolutionize agricultural practices. Through the judicious integration of Neutrosophic logic within multi-regression analysis, the report achieves precision in estimating crop loss prior to experimental interventions. Subsequently, in the experimental phase, transactions are fortified by blockchain technology, while the application of multi-regression analysis, honed through Reinforcement Learning, results in a tangible reduction in crop loss across diverse crop varieties. This multidimensional approach not only aims to enhance the productivity and sustainability of agricultural practices but also endeavors to imbue the agricultural landscape with resilience in the face of evolving challenges. By harnessing the combined potential of technology and innovation, the study

seeks to chart a course towards a future where agriculture thrives amidst adversities, ensuring food security and prosperity for generations to come (Dorri et al., 2019).

6 Conclusion and future perspective

The primary objective of this investigation was to leverage blockchain technology in predicting losses across multiple crops and determining optimal supply timings to facilitate secure transactions. Prior to embarking on the experimental phase, the study employed multi-regression analysis, aided by Neutrosophic logic, to estimate crop loss. Neutrosophic-RL demonstrates promising 12.5% MAE on $n = 6$ crops, suggesting viable uncertainty handling for smallholder systems. In the experimental phase, blockchain technology serves as a protective layer for transactions, while multi-regression analysis, enhanced by Reinforcement Learning, is employed to forecast crop loss across a diverse range of crops. This research endeavors to tackle prevalent challenges within the agriculture sector, encompassing issues such as single points of failure, transparency deficiencies, data privacy concerns, security vulnerabilities, exorbitant transaction costs, and trust deficits. This exploratory Smart Agriculture 5.0 framework demonstrates 25%–35% loss reduction potential and 85% transaction efficiency, warranting field validation across agro-climates.

6.1 Limitations

1. Dataset Scope: Analysis limited to six crops ($n = 720$ observations, 2022–2024) from Bhimavaram region. Multi-year, pan-India validation across 20+ crops needed for generalizability.
2. Blockchain Scalability: Current Ganache testnet implementation (1000 transactions) requires mainnet/Polygon deployment testing. Gas fees may limit transactions below ₹100 for smallholders.
3. Field Validation: Laboratory RL optimization (MAE 12.5%) lacks real-time IoT integration and farmer adoption data. Human-AI interaction patterns remain untested.

6.2 Future research

Looking to the future, the agricultural landscape holds tremendous potential for transformation as technological advancements continue to revolutionize farming practices. Building upon the groundwork laid by our study and the broader research community, several key avenues emerge for future exploration. Firstly, scalability will be paramount as agricultural operations become increasingly diverse and complex. Future research efforts could focus on developing scalable models and frameworks capable of adapting to various agricultural contexts and scales of production. Additionally, enhancing interoperability between different agricultural technologies and systems will be essential for optimizing efficiency. Researchers may explore integrating blockchain technology with other emerging technologies like IoT, AI, and remote sensing to create comprehensive, interoperable solutions for precision agriculture.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

RJ: Investigation, Methodology, Software, Writing – original draft. PR: Investigation, Supervision, Validation, Writing – review and editing. RP: Supervision, Validation, Writing – review and editing.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

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Acknowledgements

We like to express our sincere gratitude to the host institute for their support, which made this research possible. We also extend our appreciation to the scientist who worked in this area for their valuable contributions. Their expertise and support were instrumental in shaping the development and refinement of our research. Additionally, we thank the reviewers for their insightful comments and suggestions, which greatly improved the quality of this manuscript.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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